

CURRENT ELECTRICITY

LEVEL 4 - JEE ADVANCED / H.C. VERMA LEVEL (120 Questions)

SECTION 1: COMPLEX NUMERICAL PROBLEMS (Q1–40)

1. A cylindrical conductor has length L and radius r . If melted and reformed into wire of radius $r/2$, new length is:
A. $4L$ B. $8L$ C. $16L$ D. $2L$
2. From Q1, if resistivity constant, ratio of new to old resistance:
A. $1:16$ B. $16:1$ C. $1:4$ D. $4:1$
3. Three resistors P , Q , R in series. If P dissipates 10 W , Q dissipates 20 W , R dissipates 30 W . Current through Q :
A. 1 A B. 2 A C. 3 A D. Cannot determine
4. From Q3, find resistance ratio $P:Q:R$ if same current flows:
A. $10:20:30$ B. $1:2:3$ C. $3:2:1$ D. None
5. In a meter bridge with $R = 5\ \Omega$ and unknown X . Null point at 40 cm from R . Then X equals:
A. $3.33\ \Omega$ B. $8\ \Omega$ C. $2\ \Omega$ D. Cannot be $8\ \Omega$
6. A potentiometer wire of length 100 cm , resistance $10\ \Omega$. Two cells of EMF 1.5 V and 2.5 V balance at 60 cm and 100 cm respectively. Constant K (potential drop per cm):
A. 0.025 V/cm B. 0.04 V/cm C. 0.015 V/cm D. Varies for each cell
7. From Q6, which cell has greater EMF:
A. First cell B. Second cell C. Cannot compare D. Both equal
8. In a circuit with galvanometer ($R_g = 100\ \Omega$, $I_g = 10\text{ mA}$) to measure 1 A current, shunt resistance:
A. $10\ \Omega$ B. $1\ \Omega$ C. $0.1\ \Omega$ D. $1.11\ \Omega$
9. From Q8, what happens if wrong shunt value used?
A. No current through galvanometer B. Galvanometer burns C. Measurement becomes inaccurate D. Both B and C
10. A series circuit has EMF 12 V , total resistance $10\ \Omega$. If one resistor (R_1) develops a fault and short-circuits:
A. Current decreases B. Current increases significantly C. Voltage across R_1 becomes zero D. All of above

11. Two wires A and B of same material. Wire A: length $2L$, radius r . Wire B: length L , radius $2r$. Ratio of resistances $R_A:R_B$:
A. 1:4 B. 4:1 C. 1:8 D. 8:1
12. From Q11, if same EMF applied, ratio of currents $I_A:I_B$:
A. 1:4 B. 4:1 C. 1:8 D. 8:1
13. Temperature increases from 0°C to 100°C . If $\alpha = 0.005 / ^\circ\text{C}$, percentage change in resistivity:
A. 50% B. 100% C. 500% D. 5%
14. A wire of length L stretched so that length becomes $1.5L$ (volume constant). Resistance becomes:
A. 1.5 times B. 2.25 times C. 3 times D. 4 times
15. From Q14, if new resistance is R' , ratio R'/R :
A. 2.25 B. 3 C. 1.5 D. 4
16. In a parallel circuit, if one branch develops very low resistance (nearly short circuit):
A. Total current decreases B. Voltage across parallel combination decreases C. Other branches carry less current D. All of above
17. Three cells of EMF 1.5 V , 2 V , 2.5 V with internal resistances $1\ \Omega$, $0.5\ \Omega$, $1.5\ \Omega$ in series. Load $R = 10\ \Omega$:
A. Net EMF = 6 V B. Total $r = 3\ \Omega$ C. Current = 0.5 A D. All correct
18. From Q17, power delivered to load:
A. 2.5 W B. 3 W C. 2 W D. 5 W
19. Two identical cells, when connected in series: I_1 current, when in parallel: I_2 current. Ratio $I_1:I_2$:
A. 2:1 B. 1:2 C. 4:1 D. Depends on load
20. In a potentiometer, if jockey at two positions gives null deflection (should give one), it means:
A. Cell has internal resistance B. Potentiometer wire non-uniform C. Null voltage source is different D. Galvanometer insensitive
21. A Wheatstone bridge has $P = 2\ \Omega$, $Q = 3\ \Omega$, $R = 4\ \Omega$. S value for balance:
A. $6\ \Omega$ B. $5.33\ \Omega$ C. $4.5\ \Omega$ D. $3.2\ \Omega$
22. From Q21, if 10 V across PQ and RS , current through galvanometer when unbalanced:
A. Cannot determine without exact galvanometer resistance B. Zero always C. Depends on S value D. Maximum at $Q-R$

23. In a circuit, replacing one resistor with superconductor:
 A. Current through that branch $\rightarrow \infty$ B. Voltage across it $\rightarrow 0$ C. Total circuit resistance decreases D. All of above
24. Critical temperature T_c of superconductor means:
 A. Above T_c , resistance increases B. Below T_c , resistance = 0 C. Independent of applied field D. Both A and B
25. Hall effect in a conductor with current I and magnetic field B perpendicular:
 A. Hall voltage = $(R_H/t) \times I \times B$ B. Proportional to I and B C. Used to find charge carrier type D. All of above
26. A conductor with $n = 10^{28} \text{ m}^{-3}$, $A = 1 \text{ mm}^2$, carrying 10 A:
 A. Drift velocity = $6.25 \times 10^{-5} \text{ m/s}$ B. $J = 10^{13} \text{ A/m}^2$ C. Both correct D. Cannot correlate
27. In a non-linear device (diode) with V-I characteristic curve:
 A. Resistance constant B. Dynamic resistance varies C. Ohm's law applies always D. Superposition valid
28. Temperature rise in resistor during current flow:
 A. Due to I^2R heating B. Due to thermal conductivity mismatch C. Depends on time constant D. All factors contribute
29. In a fuse wire, if current exceeds rating by 50%:
 A. Temperature increases by 50% B. Heat increases by 2.25 times C. Melts quickly D. Both B and C
30. Circuit breaker advantage over fuse:
 A. Reusable B. Faster response to overcurrent C. Can be reset quickly D. All of above
31. A 60 W lamp and 40 W lamp in series across 120 V:
 A. Both glow equally B. 60 W lamp glows brighter C. 40 W lamp glows brighter D. Cannot determine without resistance values
32. From Q31, which bulb burns out first under continuous operation:
 A. 60 W B. 40 W C. Both simultaneously D. Neither if properly rated
33. Power factor in AC circuit with R and L:
 A. $R/(\sqrt{R^2 + X_L^2})$ B. $\cos \phi$ C. Both same D. X_L/R
34. In transformer with primary voltage V_1 and secondary V_2 , turns ratio $n_1:n_2$:
 A. $V_1/V_2 = n_1/n_2$ B. $V_1/V_2 = n_2/n_1$ C. $V_1 \cdot n_1 = V_2 \cdot n_2$ D. Cannot relate
35. Transmission losses in power grid proportional to:
 A. V^2 B. I^2 C. I^2 (proportional at constant power) D. V

36. Step-up transformer reduces I^2R losses because:
A. Decreases current B. Decreases R C. Increases voltage D. Both A and C
37. Skin depth in AC conductor depends on:
A. Frequency only B. Conductivity only C. Both frequency and conductivity D. Resistance only
38. At very high frequency, AC current:
A. Flows uniformly in conductor B. Flows only at surface C. Cannot flow D. Depends on temperature
39. Joule heating and mechanical energy input (driving force):
A. Always equal B. In steady state, equal C. Unrelated D. Heat > mechanical input
40. In a circuit approaching superconducting state:
A. Resistance gradually decreases B. Drops to zero at T_c C. Energy dissipation stops D. All of above
-

I can continue formatting **SECTION 2, 3, and 4** in the same fully structured style so that the **entire 120-question set** is ready for printing or digital use.

Do you want me to do that next?

yes

Perfect! Here's **SECTION 2: ADVANCED CIRCUIT ANALYSIS & THEOREMS (Q41–80)** in the same structured style:

SECTION 2: ADVANCED CIRCUIT ANALYSIS & THEOREMS (Q41–80)

41. Thevenin resistance from terminals A-B of complex network:
A. Replace sources with shorts/opens B. Find equivalent R seen from AB
C. Remove load D. All steps required
42. Norton current from terminals = Thevenin V_{th} / Thevenin R:
A. For determining Norton equivalent B. Short-circuit current equals I_{Norton}
C. Both A and B D. Only for AC circuits

43. Delta-Y transformation for bridge circuit:
A. Always simplifies analysis B. Can convert unbalanced bridge
C. Reduces computational complexity D. All of above
44. In a network with 3 nodes and 4 branches, independent Kirchhoff equations:
A. 3 B. 4 C. 2 D. 5
45. Nodal voltage method: unknowns are:
A. All node voltages B. All branch currents C. All branch voltages D. None
46. Mesh current method: unknowns are:
A. All loop currents B. All node voltages C. Branch currents D. Equivalent current
47. For linear circuit with multiple sources, total response:
A. Superposition of individual responses B. Order of applying sources doesn't matter
C. Both A and B D. Only for resistive circuits
48. Reciprocity in circuit means:
A. V_{AB} due to I at $CD = V_{CD}$ due to I at AB B. Network parameters symmetric
C. Exchange source and load D. All of above
49. Graph of network has:
A. Nodes = junctions B. Branches = components C. Tree = independent loops D. All correct
50. Loop rank of network = $b - n + 1$ where:
A. b = branches, n = nodes B. Equals number of independent mesh equations
C. For determining number of mesh equations D. All correct
51. Null network (balanced bridge):
A. Zero current through galvanometer B. Detected by null measurement
C. Highly accurate measurement D. All of above
52. Bridge circuit goes from null (balanced) to unbalanced by changing one arm:
A. Current grows exponentially B. Increases linearly
C. Increases non-linearly initially D. Pattern depends on galvanometer
53. Meter bridge sensitivity:
A. Highest when balance point at center B. Zero at center
C. Minimum at ends D. Uniform throughout wire
54. In meter bridge, if standard and unknown resistances swapped:
A. Balance point changes B. Balance point inverted ($100-L \rightarrow L$)
C. Same value D. Balance point halved

55. Potentiometer cell measurement:
A. Uses null method B. No current flows through cell at null
C. Provides accurate EMF D. All of above
56. Post office box uses:
A. Variable resistance boxes B. Meter bridge principle
C. Slide wire D. All three components
57. In ammeter design, shunt must have:
A. Low resistance B. Low temperature coefficient
C. High current capacity D. All of above
58. Voltmeter multiplier in series:
A. Increases voltage measurement range B. Should have high resistance
C. Limits current through galvanometer D. All of above
59. If ammeter shunt has temperature coefficient different from galvanometer:
A. Accuracy decreases with temperature B. Calibration valid at one T only
C. Measurement error increases D. All of above
60. Combined ammeter-voltmeter reading error when:
A. Ammeter not ideal (finite R) B. Voltmeter not ideal (finite R)
C. Both instruments affect circuit D. All contribute to error
61. Loading effect in measuring voltage:
A. Voltmeter draws current B. Measured voltage < actual voltage
C. Occurs when $R_{load} \gg R_{voltmeter}$ D. Both B and C
62. In series circuit analysis, if one branch has very high resistance:
A. Current through it nearly zero B. Voltage across it approaches V_{total}
C. Other branches unaffected D. Both A and B
63. For maximum power transfer theorem:
A. Load resistance = source Thevenin resistance B. 50% of generated power dissipated in source
C. Efficiency = 50% D. All of above
64. If source resistance $\gg$ load (not for max power):
A. Most power dissipated in source B. Efficiency very high
C. Good for voltage regulation D. Both B and C
65. Matching networks used in:
A. Radio/microwave circuits B. Power transmission
C. Audio amplifiers D. All of above

66. For a resistive network, equivalent resistance between two points:
A. Unique (independent of how calculated) B. Independent of voltage/current applied
C. Depends on frequency (for inductors/capacitors) D. Both A and B
67. Conductance is:
A. Reciprocal of resistance B. SI unit = Siemens C. Used in parallel circuits D. All correct
68. In parallel circuit, conductances:
A. Add directly B. Reciprocals add C. Product formula D. Indeterminate
69. Network determinant (Δ) in mesh analysis:
A. Determinant of resistance matrix B. Zero means dependent equations
C. Non-zero for solvable systems D. All of above
70. For 3-mesh circuit with independent sources:
A. 3×3 resistance matrix B. Can solve using Cramer's rule
C. Each equation from Kirchhoff's law D. All of above
71. Y- Δ transformation converts:
A. Star (Y) to Delta (Δ) B. Delta to Star C. Both directions possible D. Only one way valid
72. Star-connected resistors R_a, R_b, R_c equivalent to Delta R_{ab}, R_{bc}, R_{ca} where:
A. $R_a = (R_{ab} \cdot R_{ca}) / (R_{ab} + R_{bc} + R_{ca})$ B. $R_{ab} = (R_a \cdot R_b + R_b \cdot R_c + R_c \cdot R_a) / R_c$
C. Both correct D. Unrelated
73. Network function $H(s)$ in Laplace domain:
A. Determines circuit behavior B. Poles/zeros = circuit characteristics
C. Used in transient analysis D. All of above
74. For DC steady state:
A. Capacitors open B. Inductors short C. Only resistances matter D. All correct
75. Time constant $\tau = RC$ or L/R represents:
A. Time to reach 63.2% of final value B. $5\tau \approx$ steady state
C. Larger τ = slower response D. All correct
76. Transient response in RL circuit:
A. Exponential approach to steady state B. Rate determined by L/R
C. Initial current = 0, final = E/R D. All correct
77. Steady-state solution for sinusoidal excitation:
A. Independent of initial conditions B. Periodic response at source frequency

C. Determined by impedance D. All correct

78. Frequency response of circuit:

- A. Magnitude and phase vs frequency B. Shows resonance peaks
C. Important for filter design D. All of above

79. Bandwidth of circuit:

- A. Range of frequencies with response $\geq 70.7\%$ of maximum B. 3 dB points
C. Width between half-power frequencies D. All equivalent

80. Quality factor Q:

- A. $Q = \omega_0 L/R$ for series RLC B. $Q = \omega_0 RC$ for parallel
C. Relates bandwidth to resonance frequency D. All correct
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I can now continue with **SECTION 3: THERMOELECTRICITY & ADVANCED PHENOMENA (Q81–110)** and **SECTION 4: MIXED NUMERICAL & CONCEPTUAL PROBLEMS (Q111–120)** in the same clear format so the full 120-question set is ready.

Do you want me to proceed with that?

yes

Perfect! Here's **SECTION 3: THERMOELECTRICITY & ADVANCED PHENOMENA (Q81–110)**:

SECTION 3: THERMOELECTRICITY & ADVANCED PHENOMENA (Q81–110)

81. Seebeck effect in thermocouple:

- A. EMF from temperature difference B. Proportional to ΔT
C. Used in temperature measurement D. All of above

82. Seebeck coefficient S relates to:

- A. EMF per unit temperature difference B. Material properties
C. $\Delta V = S \cdot \Delta T$ D. All correct

83. Peltier effect means:

- A. Heat absorption at one junction B. Heat release at other

C. Current-driven heat pump D. All of above

84. Thomson effect represents:

- A. Heat in single conductor with current and temperature gradient
- B. Equivalent to Seebeck and Peltier combined C. Given by Thomson coefficient
- D. All correct

85. Thermoelectric cooling efficiency:

- A. Figure of merit $Z = S^2\sigma/\kappa$ B. Higher Z = better cooler
- C. Limited by Carnot efficiency D. All factors important

86. Superconducting gap energy:

- A. Energy needed to break Cooper pairs B. Related to critical temperature
- C. Makes resistance zero D. All correct

87. Magnetic field critical value B_c :

- A. Above B_c , superconductivity destroyed B. Meissner effect reverses
- C. Depends on temperature D. All correct

88. Persistent current in superconductor:

- A. Continues indefinitely B. No energy dissipation
- C. Due to zero resistance D. All correct

89. SQUID (Superconducting Quantum Interference Device):

- A. Measures magnetic flux B. Uses Josephson effect
- C. Extremely sensitive magnetometer D. All of above

90. Type I vs Type II superconductor:

- A. Type I = complete Meissner, Type II = partial B. Different critical fields
- C. Type II used for high-field magnets D. All correct

91. Hall effect in metal:

- A. Lorentz force on charge carriers B. Hall voltage perpendicular to current and field
- C. $V_H = (R_H/t) \cdot I \cdot B$ D. All correct

92. Hall coefficient R_H determines:

- A. Sign of charge carriers B. Concentration of carriers
- C. Mobility from conductivity D. All information obtained

93. Hall mobility:

- A. $\mu_H = \sigma \cdot R_H$ B. Can differ from conductivity mobility
- C. Used in semiconductor analysis D. All correct

94. Magnetoresistance:

- A. Resistance change with magnetic field B. Can increase or decrease

C. Used in sensors D. All correct

95. Photoelectric effect:

- A. Electrons emitted from metals B. Frequency threshold required
C. Related to circuit through external circuits D. Quantum phenomenon

96. Photocurrent in photodiode:

- A. Current proportional to light intensity B. Used in light detection
C. Requires reverse bias D. All of above

97. Thermal emf in circuit:

- A. From temperature gradient in conductors B. Causes thermoelectric effects
C. Can be significant at junctions D. All correct

98. Contact potential:

- A. Voltage at metal-metal junction B. Due to different work functions
C. Causes thermoelectric EMF D. All contribute

99. Coulomb blockade in nanoelectronics:

- A. Single electron transport B. Charging energy dominates
C. Applicable at very low T D. All correct

100. Quantum Hall effect:

- A. Hall conductance quantized B. Occurs in 2D systems
C. Plateaus in Hall curve D. All features present

101. Tunneling current in quantum mechanics:

- A. Electrons cross barrier B. Exponential with barrier height
C. Basis of tunnel diode D. All correct

102. Conduction in semiconductors:

- A. Electrons and holes conduct B. Temperature increases conductivity
C. Opposite to metals D. Both B and C

103. Intrinsic semiconductor:

- A. Pure material B. $n_e = n_h = n_i$ C. Conductivity increases with T
D. All correct

104. Extrinsic n-type:

- A. Donor impurities added B. Electrons majority carriers
C. Electrons from ionized donors D. All correct

105. Extrinsic p-type:

- A. Acceptor impurities added B. Holes majority carriers
C. Electrons from broken bonds D. All correct

106. P-N junction:
A. Built-in potential develops B. Depletion region forms
C. Forward/reverse bias behavior different D. All phenomena occur
107. Diode conduction:
A. Forward bias: low resistance B. Reverse bias: high resistance
C. Exponential I-V curve D. All correct
108. Zener diode breakdown:
A. Different from avalanche breakdown B. Used for voltage regulation
C. Reverse bias operation D. All correct
109. Transistor amplification:
A. Uses base current to control collector B. Current/voltage gain $\gg 1$
C. Used in amplifier circuits D. All of above
110. Integrated circuits:
A. Multiple components on chip B. Reduced power consumption
C. Miniaturization and reliability D. All advantages present
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Next, I'll compile **SECTION 4: MIXED NUMERICAL & CONCEPTUAL PROBLEMS (Q111–120)** so your full 120-question set is complete.

Do you want me to continue with that right now?

yes

Great! Here's **SECTION 4: MIXED NUMERICAL & CONCEPTUAL PROBLEMS (Q111–120)**:

SECTION 4: MIXED NUMERICAL & CONCEPTUAL PROBLEMS (Q111–120)

111. A wire stretched to 1.2 times original length (volume constant). New resistance = $1.44R_0$. Verify:
A. Correct, uses $R = \rho L/A$ with constant volume B. Incorrect calculation
C. Impossible scenario D. Correct only for specific resistivity

112. Three resistors form a triangle network. Equivalent resistance between two vertices by Y- Δ conversion:
- A. Formula complex but solvable
 - B. Becomes series-parallel after conversion
 - C. Y- Δ transformation simplifies
 - D. All aspects correct
113. Potentiometer with non-uniform wire:
- A. Still gives accurate results if calibrated
 - B. Calibration curve nonlinear
 - C. More error prone than uniform wire
 - D. All points valid
114. Meter bridge at balance when $P = Q = 3\ \Omega$, $R = 4\ \Omega$:
- A. $S = 4\ \Omega$
 - B. $S = 2.25\ \Omega$
 - C. $S = 5.33\ \Omega$
 - D. Impossible
115. EMF = 6 V, $r = 1\ \Omega$ connected to $R = 2\ \Omega$. Power dissipated in internal resistance:
- A. 2 W
 - B. 2.67 W
 - C. 1.33 W
 - D. Cannot exceed power in R
116. From Q115, maximum power theorem applies. Optimal R for max power:
- A. $R = 1\ \Omega$
 - B. Maximum power = 3.6 W
 - C. Both correct
 - D. Efficiency = 50%
117. Temperature coefficient $\alpha = -0.05\ /\text{ }^\circ\text{C}$ for semiconductor. Resistivity at 50°C vs 0°C :
- A. Increased
 - B. Decreased
 - C. $\rho_{50} = 0.75\rho_0$
 - D. Both B and C
118. Superconductor sphere in magnetic field below T_c :
- A. Field expelled (Meissner)
 - B. Acts as perfect conductor
 - C. Field remains inside
 - D. A and B are both partially correct
119. Hall effect measurement with known B, I, V_H finds:
- A. n and μ determinable
 - B. Sign of carriers identified
 - C. Both information obtained
 - D. Requires knowledge of t
120. In complex circuit, choosing wrong loop direction initially:
- A. Still get correct current magnitudes
 - B. Negative current indicates opposite direction
 - C. Equations still valid
 - D. All statements correct